Konux: Railway Switches or Several Industries?

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MODULE 03 · THE PROBLEM SPACE · CASE, READ BEFORE CLASS

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At the end of October 2016, Andreas Kunze, co-founder and chief executive of the Munich start-up Konux, led a company that was pursuing two directions at once. Konux could build itself around the railway switches of Deutsche Bahn (DB) and other railways, or it could keep developing its sensors and analysis software for several industries. Its homepage listed railway, industrial pumps, machine tools and industrial automation.[1] On 19 October 2016 Konux had been one of ten winners of the German Mobility Prize, awarded by the Federal Ministry of Transport and Digital Infrastructure and the initiative "Deutschland – Land der Ideen", for a project that monitored railway infrastructure for DB with sensors and data analysis based on artificial intelligence.[2]

A few days before the award, on 13 October, Max Hasler, Konux's vice president of finance, had described the company's position in a guest lecture by video to students at Western New Mexico University. In rail, he said, Konux had a very big reference customer in DB and was negotiating a first roll-out on the most critical switches. In parallel it ran projects in machine tools and industrial pumps and was talking to elevator companies. The money from the last financing round would probably last until the first quarter of 2018, and the next round was planned for the beginning of 2018. Until then Konux would focus on bringing DB to a large roll-out, and would see at the next round how its products developed.[3]

Kunze led a management team of five and a company of 35 employees.[4] [5] Whichever direction Konux chose would have to carry it until the next financing round.

Switches at Deutsche Bahn

A railway switch moves a short section of track so that a train can pass from one track onto another. At the end of 2015 DB Netz AG, the DB subsidiary that ran most of Germany's rail network, counted 67,071 switches and crossings, down from 71,674 in 2010. Their average age was 19.5 years, and DB Netz put their technical life at 20 to 42 years, 34 years on average (Exhibit 1).[6] Infrastructure owned by DB carried almost 98 percent of all train operations in Germany in 2014, although about 160 companies operated rail infrastructure in total.[7]

In 2015 DB Netz logged 187,338 reports of faults with operational impact that it or its contractors had caused, 33,108 of them at switches and crossings. Faults at switches and crossings of the two most urgent priorities stayed open for 1,725,246 minutes in total, out of 9,704,178 minutes for all asset groups.[8] DB recorded 174.6 million minutes of train delay in 2015 from all causes, up from 162.4 million in 2014 (Exhibit 1).[9] The reports did not show how many of these delay minutes switch faults had caused.

DB Netz checked its assets in periodic inspections under its own rulebook. In some areas it also monitored them technically, switches for example with diagnosis equipment.[10] Inspecting the critical parts of a switch had otherwise been possible only by hand, Munich Startup reported in October 2016.[2]

DB's published maintenance figures did not single out switches. In 2014 DB's infrastructure companies spent EUR 1.65 billion on maintenance of the existing network under their funding agreement with the federal government, the Leistungs- und Finanzierungsvereinbarung (LuFV), and EUR 3.31 billion on replacement investment. Maintenance spending rose to about EUR 1.72 billion in 2015. Both figures covered all asset groups, from track and bridges to signalling.[11] The second LuFV guaranteed EUR 28 billion for the years 2015 to 2019.[12] For switches alone, DB Netz renewed 1,883 in 2015 and planned EUR 1,771 million of switch investment for 2016 to 2020, about 9,500 renewals. Of that sum, EUR 1,640 million fell in the asset classes covered by the LuFV, an average of EUR 328 million a year.[13] The reports did not state what inspecting switches cost.

Deutsche Bahn and Digital Diagnosis

DB says it opened the mindbox, a programme for start-ups, in Berlin in 2015.[14] In June 2015 DB chose the first four start-ups for a new accelerator programme. Each received EUR 25,000, office space and access to DB's data for three months, and DB would continue the work with those whose projects proved themselves. Konux was one of the four. Golem wrote that Konux wanted to use its own sensor technology to monitor the condition of rail infrastructure in real time and make preventive maintenance possible.[15]

In November 2015 DB's chief technology officer, Josef Stoll, described six digitalisation initiatives to the magazine CIO. One of them, "Infrastruktur 4.0", aimed to raise the utilisation of the network by 20 to 40 percent with big data, for example through better maintenance. "In this way we can in future analyse a switch's power consumption and see when a fault is coming," Stoll said.[16]

DB's annual report for 2015, dated February 2016, said that digital switch diagnosis should reduce failures by up to 50 percent.[17] In April 2016 DB Netz wrote in its infrastructure report that it would equip 30,000 switches with sensors for automatic condition diagnosis over the medium term. By continuously measuring the forces at the drive that moves a switch, it expected to detect many faults in the switch and the drive before they caused delays. Maintenance at fixed intervals was to give way to maintenance based on the condition of each switch and on fault forecasts. In 2015 DB Netz had classified all its switches by operational importance, and it was further developing pilots for the automatic inspection of switches (Exhibit 4).[18]

Sensors for Industry

Konux was not the only company that sold sensors to industry. In April 2016 Handelsblatt named established mid-sized makers such as ifm electronic, Pepperl+Fuchs and HBM. Kunze told the paper that Konux set itself apart through the intelligence of its software, since its customers received continuous data analysis on top of the sensors. Peter Frühauf of the VDMA, the German association

of machinery makers, answered that other suppliers could do that too. He added that the established sensor makers were “certainly not as agile” as a start-up and that demand for sensors had been rising for years.[19]

Surveys of the time described a large industrial market that had only begun to digitalise. In 2014 the consultancies PwC and Strategy& surveyed 235 German industrial companies. Nearly a fifth had digitalised the key processes of their value chain. PwC extrapolated that German industry would invest EUR 40 billion a year until 2020 in “Industrie 4.0” solutions, the German label for digitally networked industrial production, and expected 85 percent of companies to have implemented them in all important business areas within five years.[20] In October 2015 a self-assessment of German machinery makers published by the VDMA’s IMPULS foundation counted 5.6 percent of them as pioneers of Industrie 4.0, 17.9 percent as beginners and 76.5 percent as newcomers.[21] Konux’s own statements and the press reports cited in this case give no figures for the markets in pumps, machine tools or elevators.

Konux, 2014 to 2016

Kunze studied information systems at the Technical University of Munich (TUM) and took part in Manage&More, an entrepreneurship programme at TUM.[22] [23] Konux had existed as a company since March 2014.[15] Handelsblatt wrote in 2016 that Kunze had built it with four colleagues who studied at TUM and at Ludwig Maximilian University (LMU) in Munich.[19] Its founding team included Kunze, Vlad Lata and Dennis Humhal. Sources from 2015 and 2016 differ on who else belonged to it.[24]

Konux began with a measurement technology. Computerwoche wrote in July 2014 that the company had spent several years developing an optical, contactless technology that could measure many different quantities, and that it pursued “the vision of revolutionising the sensor market for mechanical measurands”.[23] In mid-June 2014 Konux was among the ten winners of the World Founder Forum, a competition of the Californian investor Founder.org with 50 finalists, and received USD 100,000.[23] In 2014 it also joined the start-up accelerator of EIT Digital, an innovation network initiated by the European Institute of Innovation and Technology, after winning the first prize in the cyber-physical systems category of EIT Digital’s Idea Challenge.[25]

In spring 2015 Konux raised its first financing round. Gründerszene reported in May 2015 that Michael Baum, the founding chief executive of the software company Splunk, led the round of USD 1.6 million through his programme Founder.org, together with German business angels. Hasler put the round at 2 million in April 2015, without naming a currency, and Kunze recalled it in 2017 as 2 million dollars.[26] Kunze said in 2017 that he doubted anyone in Germany would have given the student founders that money at that valuation, when they did not know exactly which market they would enter and had an idea about a technology but perhaps no proof of concept yet.[27] Hasler told the students in October 2016 that roughly 500,000 of that first money had been invested very well and that the rest had gone into very expensive failures, which he saw as usual for a start-up.[28]

In July 2015 Munich Startup described Konux as a maker of individualised sensor solutions based on its patented optoelectronic measurement principle. The principle worked in nine sensor types that measured quantities such as torque, pressure, force or angle. Konux had grown from 4 to about 30 employees within a year.[22] Kunze’s advice to other founders on winning customers was to know what customers want and above all what they need, to offer the solution to a problem, and to research carefully before the first contact.[22] In August 2015 Lata, Konux’s chief technology officer, told the trade magazine A&D that besides screw automation Konux was approaching collaborative robots, was represented in agricultural machinery and saw test stands as an important market. Software would

come next.[29] In October 2015 Konux's German homepage carried a quote from Klaus Müller, DB's chief technology officer for infrastructure, about Konux's role on the way to "Infrastruktur 4.0", above its nine sensor types.[30] In March 2016 its English homepage listed torque, angle and position sensors and an accelerometer, with applications from test stands and industrial automation to tractor power take-offs, screw systems, extruders, bottling, drilling and robotics. Rail was not among them.[31]

On 14 April 2016 Konux announced a USD 7.5 million Series A round led by MIG, a German venture capital firm, with UnternehmerTUM, New Enterprise Associates (NEA) of Silicon Valley and the investor Andy Bechtolsheim. The money was to expand the product platform, roll out products in volume for new and existing customers and "establish KONUX as the leading company for machine insights". Kunze said in the announcement, "We believe the real impact and the toughest challenges of IoT [the Internet of Things] are in the industrial world." Konux promised its customers maintenance cost reductions of "25%+" and called DB a "major customer". Müller of DB said, "KONUX is one partner on our essential road to infrastructure 4.0." Greg Papadopoulos of NEA said that European railway track companies alone spent USD 25 billion a year on infrastructure maintenance.[32] EIT Digital's report of the round named railway, aviation, pipelines, robotics and industrial automation as fields of application.[25]

Two weeks later Handelsblatt reported that Kunze was equipping part of DB's 70,000 switches with sensors. Konux employed 25 people and had earned revenue of just under EUR 100,000 ("einen knapp sechsstelligen Umsatz") in its last financial year. "We want to be market leader in the rail network segment in three to five years," Kunze said. He put the number of switches at about 350,000 in Europe and more than a million worldwide. Handelsblatt added that Konux was thinking further, to industrial pumps, machine and plant engineering and medical technology.[33]

In July 2016 Konux's company page said that since 2015 cooperation projects with partners such as DB had gathered possible uses for its sensors. They were applied, the page said, in railway infrastructure, screw automation, industrial pumps, machine tools and testing equipment.[34] In September 2016 Konux had 35 employees.[5]

What Konux Knew in October 2016

Rail. Kunze recalled in December 2016 that Konux had worked with DB inside the mindbox and had built, within three months, a solution for a problem in DB's infrastructure, a sensor that captured the impact of passing trains on the switch.[35] In March 2016 Lata said that the DB project aimed to make work safer for the staff who had to go out to the track, through continuous monitoring that helped DB predict its maintenance. Konux had sensors deployed in the field, was gathering first data and was looking at what in that data would benefit the customer.[36]

Hasler described the rail product in October 2016. Konux put a sensor on the switch, which he called the most expensive asset in the rail network. For DB, he said, delay minutes were a big and costly problem. The sensor contained low-cost vibration sensors, sent its data over the mobile phone network and ran on a battery, so that it did not have to be connected to the existing power supply, which would have made the application far more safety-critical. Sensors stood at field test locations, and Konux was negotiating a first roll-out on the most critical switches. The aim was all 70,000 switches in Germany.[37] Hasler put the possible saving at about EUR 13,000 per switch per year, a figure he called a very basic calculation.[38] Munich Startup reported on 24 October 2016 that DB already used Konux's solution to monitor switches in real time.[2]

Hasler also described how DB had measured a switch before. A maintenance crew installed measuring equipment by hand, waited for trains to pass, took notes, took the equipment down and moved it to the next switch.[39] Konux's page for the German Mobility Prize, which carried an interview with Kunze and gave the staff count as of September 2016, said its system replaced "current manual measurement methods" with sensors on the track.[5] Looking back in 2017, Kunze said that the work with DB meant learning from people who had, for example, gone out every day for 30 years and could hear from the sound of a passing train how well the frog, the centre piece of a switch, was doing.[40]

In the interview on that page, Kunze said that Konux's starting point had been that companies could profit above all from precise and intelligently networked sensors. Konux had then realised that processing and analysing the data in real time gave customers even more value, and now offered sensors, "if needed", together with analysis based on artificial intelligence. Within two years Konux wanted to see its solutions on railway tracks in more European countries and in the USA. "We also want to expand our industrial fields of application further," he said, naming industrial automation.[5]

In August 2016 Kunze spoke on the podcast Startuperfolg. Its show notes list chapters on how Konux had understood its customers' needs "mit 1892 Telefonaten" (with 1,892 phone calls), on the target groups of its user interviews, on buyer personas and on how a customer had led Konux to its software component. The show notes do not say what the calls found.[41]

Other industries. Hasler said in October 2016 that Konux had developed minimum viable products for pumps and for machine tools and had adapted its sensor to fit on pumps, which he called a very expensive asset as well. Konux was doing projects in machine tools and industrial pumps and talking to elevator companies, because, he said, these industries were not very digitalised and could all profit a lot from its solution.[42]

The terms of Konux's work with DB, the size of the roll-out under negotiation, the results of its projects outside rail and its revenue after 2015 were not publicly disclosed.

Two Directions

Build around rail. This direction followed Kunze's stated aim of April 2016 to lead the rail segment within three to five years and Hasler's plan of October 2016 to bring DB to a large roll-out before the next round. DB had worked with Konux since its accelerator of 2015, its chief technology officer for infrastructure had spoken for Konux in public, and it had announced sensors for 30,000 switches and up to 50 percent fewer failures from digital diagnosis. Switches and crossings drew tens of thousands of fault reports a year, and DB had measured switches by hand. Against it stood other facts in the record. Infrastructure owned by DB carried almost 98 percent of train operations in Germany. DB already monitored some switches with diagnosis equipment and planned its own diagnosis at the switch drive. The roll-out was still under negotiation, and Konux's revenue in its last financial year had been just under EUR 100,000.

Keep several industries. This direction followed the Series A, raised to build "the leading company for machine insights", Kunze's statement that the toughest challenges of the Internet of Things lay in the industrial world, and his plan to expand industrial applications such as automation. Konux had built products for pumps and machine tools, was talking to elevator companies and had served screw automation with its earlier optoelectronic sensors. The surveys described an industrial market that had only begun to digitalise, and the VDMA's expert saw established sensor makers as less agile than a start-up. Against it stood that the same expert said other suppliers could also add data analysis, and that Konux's earlier industrial markets had used a different sensor from the one it now built.

A team of 35 had to carry whichever direction Konux chose.

The End of October 2016

At the end of October 2016 Konux's homepage still offered railway, industrial pumps, machine tools and industrial automation.[1] Its money was expected to last until the first quarter of 2018, and the next financing round was planned for the beginning of 2018.[3] Kunze had a large customer in one industry and projects or talks in several others. Where should Konux put its people and its money?

Exhibit 1: Switches at Deutsche Bahn, 2010 to 2015

	2010	2011	2012	2013	2014	2015
Switches and crossings, DB Netz	71,674	70,477	69,983	69,400	68,516	67,071
Average age of switches, years	19.6	19.6	19.9	20.0	19.9	19.5
Fault reports, all asset groups (priorities 1 to 4)	205,758	195,514	199,953	194,914	183,890	187,338
of which at switches and crossings	36,338	33,025	35,555	35,854	32,152	33,108
Fault duration, all asset groups, minutes (priorities 1 and 2)	8,180,770	7,903,838	8,272,573	9,366,058	9,210,786	9,704,178
of which at switches and crossings	1,377,288	1,309,155	1,449,744	1,630,366	1,604,833	1,725,246
Train delay, all causes, million minutes	159.12	155.48	146.52	164.02	162.43	174.63

Spending	Amount
Maintenance of the existing network under the LuFV, all asset groups, 2014	EUR 1.650 billion
Maintenance of the existing network under the LuFV, all asset groups, 2015	about EUR 1.72 billion
Switches renewed by DB Netz, 2015	1,883
Switch investment planned by DB Netz for 2016 to 2020	EUR 1,771 million, about 9,500 renewals
of which in the asset classes covered by the LuFV	EUR 1,640 million, EUR 328 million a year on average

Fault reports count faults of priorities 1 to 4, the priorities with operational impact out of seven, caused by DB Netz or its contractors at bridges, tunnels, level crossings, tracks, switches and crossings and signalling. Fault duration runs from the report to the release of the asset and counts priorities 1 and 2. The maintenance figures cover all of DB's infrastructure companies and all asset groups, not switches alone.

Source: compiled by the case writer from DB Netz AG, Infrastrukturzustands- und -entwicklungsbericht 2014 and 2015.[6 11 8 9 13 43]

Exhibit 2: Konux, March 2014 to October 2016

Date	Event	Reported by
March 2014	Konux exists as a company in Munich	Golem
June 2014	Winner at the World Founder Forum, USD 100,000. Vision of an optical sensor technology for mechanical quantities	Computerwoche
2014	Joins EIT Digital's accelerator after a first prize in its Idea Challenge	EIT Digital
Spring 2015	First financing round, USD 1.6 million (May 2015) or 2 million (Hasler, 2016, and Kunze, 2017)	Gründerszene, Hasler, Kunze
June 2015	One of the first four start-ups in DB's new accelerator, three months, EUR 25,000 each	Golem
July 2015	About 30 employees, nine sensor types on one optoelectronic principle	Munich Startup
August 2015	Markets named are screw automation, collaborative robots, agricultural machinery and test stands	Lata (Konux)
October 2015	German homepage quotes DB's chief technology officer for infrastructure	Konux
March 2016	English homepage lists sensors and industrial applications, no rail	Konux
14 April 2016	Series A, USD 7.5 million, to become "the leading company for machine insights"	Konux
April 2016	25 employees, revenue in the last financial year just under EUR 100,000, part of DB's switches being equipped	Handelsblatt
July 2016	Company page lists railway infrastructure, screw automation, industrial pumps, machine tools and testing equipment	Konux
September 2016	35 employees	Konux
13 October 2016	DB pilot, roll-out under negotiation, products for pumps and machine tools, money until early 2018	Hasler (Konux)
19 October 2016	German Mobility Prize for the rail project	Munich Startup
30 October 2016	Homepage lists Railway, Industrial Pumps, Machine Tools and Industrial Automation	Konux

Source: compiled by the case writer from the sources in notes [15], [23], [25], [26], [22], [29], [30], [31], [32], [33], [34], [5], [3], [2] and [1].

Exhibit 3: What Konux Said About Its Markets, 2015 to 2016

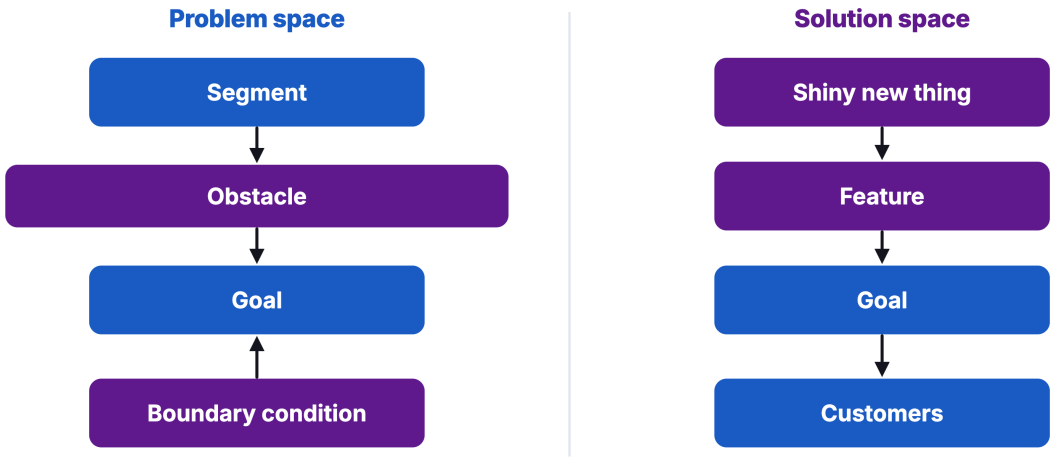
Date	Who, where	What was said
August 2015	Lata, A&D	Besides screw automation, Konux was approaching collaborative robots, was represented in agricultural machinery and saw test stands as an important market
April 2016	Kunze, Series A release	"We believe the real impact and the toughest challenges of IoT are in the industrial world." Customers would cut maintenance cost by "25%+"
April 2016	Kunze, Handelsblatt	"We want to be market leader in the rail network segment in three to five years." About 350,000 switches in Europe, more than a million worldwide
April 2016	Handelsblatt	Konux was thinking further, to industrial pumps, machine and plant engineering and medical technology
July 2016	Konux company page	Sensors applied in railway infrastructure, screw automation, industrial pumps, machine tools and testing equipment
2016, undated (staff as of September)	Kunze, Mobility Prize page	Within two years on railway tracks in more European countries and the USA. "We also want to expand our industrial fields of application further"
13 October 2016	Hasler, lecture (paraphrase)	DB a very big reference customer, roll-out under negotiation. Projects in machine tools and pumps, talks with elevator companies, because these industries were not very digitalised
30 October 2016	Konux homepage	Sensor solutions "for industrial companies", with Railway, Industrial Pumps, Machine Tools and Industrial Automation as applications

Source: compiled by the case writer from the sources in notes [29], [32], [33], [34], [5], [42] and [1]. German statements translated by the case writer.

Exhibit 4: Deutsche Bahn on Switch Diagnosis, 2015 to 2016

Date	Who, where	What was said
June 2015	DB's first start-up round, as reported by Golem	Four start-ups chosen, among them Konux, which wanted to monitor rail infrastructure in real time with its own sensors
October 2015	Klaus Müller, DB's chief technology officer for infrastructure, on Konux's homepage	Konux one of DB's partners on the way to "Infrastruktur 4.0"
November 2015	Josef Stoll, DB's chief technology officer, in CIO	"In this way we can in future analyse a switch's power consumption and see when a fault is coming"
February 2016	DB, annual report 2015	Digital switch diagnosis should reduce failures by up to 50 percent
April 2016	DB Netz, infrastructure report 2015	30,000 switches to be equipped with sensors for automatic condition diagnosis in the medium term, by measuring the forces at the switch drive. All switches classified by operational importance in 2015. Pilots for automatic switch inspection under further development

Source: compiled by the case writer from the sources in notes [15], [30], [16], [17] and [18]. German statements translated by the case writer.



Problem space and solution space, the anchor graphic of Module 03

Endnotes

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24. Gründerszene (15 May 2015) and Munich Startup (6 July 2015) named Husam Ghanem, Kunze, Lata and Humhal. A&D (14 August 2015) called Lata one of four co-founders. EIT Digital's report of April 2016 showed Humhal, Kunze and Lata as co-founders. Handelsblatt (29 April 2016) wrote that Kunze built the company with four colleagues. Hasler said in his lecture of 13 October 2016, at 37:43 to 37:58, that a founder had asked him to join. Full citations in the notes on those sources.
25. "KONUX, a startup supported by EIT Digital, raises $7.5 million", 28DIGITAL (EIT Digital), April 2016, https://28digital.eu/newsroom/news/archive/konux-a-startup-supported-by-eit-digital-raises-75-million/, accessed 2026-10-19.
26. "Warum diese Gründer im Valley nach Investoren suchen", Gründerszene, 15 May 2015, archived at https://web.archive.org/web/20150517/http://www.gruenderszene.de/allgemein/silicon-valley-finanzierung-konux-seed, accessed 2026-10-19. Hasler, lecture at Western New Mexico University, 13 October 2016, 13:02 to 13:21. Kunze, TUM School of Management panel, recording published 14 December 2017, 15:23 to 15:45. Baum as Splunk's founding chief executive from Konux's Series A release of 14 April 2016. Hasler also mentioned Andy von Bechtolsheim in connection with the first round (13:53 to 14:04) but said later in the lecture, at 53:19 to 53:34, that Bechtolsheim had not invested at first and had introduced Konux to others, so the case leaves his part in the first round open.
27. Andreas Kunze on the panel "Neue Herausforderungen in einer komplexen Unternehmenswelt", TUM School of Management, recording published 14 December 2017, YouTube, 15:23 to 16:07, https://www.youtube.com/watch?v=Ie8Yv88qbRg, accessed 2026-10-19. Reported in paraphrase from the recording, which is in German.
28. Hasler, lecture at Western New Mexico University, 13 October 2016, 13:05 to 13:38.
29. "Wir messen bis 0,01 Prozent genau", interview with Vlad Lata, A&D (industr.com), 14 August 2015, https://www.industr.com/de/wir-messen-bis-0-01-prozent-genau-252407, accessed 2026-10-19.
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31. Konux, homepage, snapshot of 19 March 2016, https://web.archive.org/web/20160319/https://www.konux.com/, accessed 2026-10-19.
32. Konux, "KONUX raises $7.5M Series A from Silicon Valley VCs", press release, 14 April 2016, https://www.konux.com/konux-raises-7-5m-series-silicon-valley-vcs/, archived at https://web.archive.org/web/20160414/https://www.konux.com/konux-raises-7-5m-series-silicon-valley-vcs/, accessed 2026-10-19.
33. Franz Hubik, "Start-up Konux: Die Defekt-Vorherseher", Handelsblatt, 29 April 2016, pages 1 and 2, http://www.handelsblatt.com/technik/hannovermesse/start-up-konux-die-defekt-vorherseher/13522016.html, archived at https://web.archive.org/web/20160430/http://www.handelsblatt.com/technik/hannovermesse/start-up-konux-die-defekt-vorherseher/13522016.html, accessed 2026-10-19. Quotes translated from German by the case writer.
34. Konux, "Company", snapshot of 7 July 2016, https://web.archive.org/web/20160707/https://www.konux.com/company/, accessed 2026-10-19.
35. Andreas Kunze on the panel "Digitale créateure – Start-ups als Treiber für Innovation und Digitalisierung", Federal Ministry for Economic Affairs and Energy, recording published 14 December 2016, YouTube, 20:08 to 20:30, https://www.youtube.com/watch?v=ALTwsTXXdi4, accessed 2026-10-19. Reported in paraphrase from the recording, which is in German.
36. Vlad Lata, "FMXA @ IOT Tech Expo – KONUX", FMXA, YouTube, 30 March 2016, 0:43 to 2:10, https://www.youtube.com/watch?v=fbo9DKWlJxU, accessed 2026-10-19. Reported in paraphrase from the recording.
37. Hasler, lecture at Western New Mexico University, 13 October 2016, 7:40 to 8:45 and 9:16 to 9:56.
38. Hasler, lecture at Western New Mexico University, 13 October 2016, 8:45 to 9:10.

39. Hasler, lecture at Western New Mexico University, 13 October 2016, 12:25 to 12:52.
40. Kunze, TUM School of Management panel, recording published 14 December 2017, 19:12 to 19:43.
41. Björn Erkens with Andreas Kunze, "22 Jahre jung, 1892 Telefonate und Venture Capital über 7,5 Millionen $", Startuperfolg Podcast, episode 17, 8 August 2016, show notes, http://startuperfolg.com/andreaskunze/, accessed 2026-10-19.
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43. DB Netz AG, *Infrastrukturzustands- und -entwicklungsbericht 2015*, p. 100, definitions of fault reports and fault duration.